

Chapter 5

Gas-Phase Method of Metal Powder Production

Irina V. Frishberg

Institute of Metallurgy, Ural Branch of the Russian Academy of Science, Yekaterinburg, Russia

Chapter Outline

Introduction	187	Powder Production and Application of Gas-Phase	
Theoretical Basis of the Gas-Phase Method	187	Processes	194
Mechanisms of Powder Formation During Vapor		Physical Vapor Deposition (PVD) Process	195
Condensation	189	Properties of Powders and Areas of Their Usage	196
Peculiarities of the Formation of Alloy Powders	193	Chemical Vapor Deposition (CVD) Process	197
Controlling Principles of the Condensation		Health and Environment Protection	198
Process	194	References	199

INTRODUCTION

The theory and technique of obtaining particulate precipitates from metals or their compound vapors are the essence of the gas-phase method [1,2]. Two types of gas-phase precipitation, physical vapor deposition (PVD) and chemical vapor deposition (CVD), are known. In the former, the powder is formed as a result of evaporation of a metal followed by condensation of its vapor; the second one involves chemical interactions (see Chapter 20). These techniques are usually called inert gas condensation (IGC) and chemical vapor reaction (CVR), respectively. The gas-phase methods are universal and sufficiently economical. They are versatile and allow the manufacturing of powders with targeted properties. The technique is applicable for producing powders from volatile metals such as zinc, cadmium, lead, magnesium, tin, copper, and their alloys as well as from a variety of metals capable of forming volatile halides, carbonyls, and metal organic compounds. Inert gas condensation methods are available for the production of nanopowders.

The process of powder formation comprises three stages:

- (1) Vaporization, either by evaporating the metal or by its chemical interaction with the components of the gas phase.
- (2) Feeding the vapor into the zone of condensation (decomposition).
- (3) Formation of solid particles by condensing or decomposing the gaseous chemical compounds (nucleation and growth of a new condensed phase).

In the case of physical condensation, the temperature of the vapor source is higher than that of the powder formation zone; in most cases of chemical precipitation it is vice versa.

THEORETICAL BASIS OF THE GAS-PHASE METHOD

Temperature (T) and pressure (p) are the principal thermodynamic characteristics of the process. The evaporation and precipitation take place at T_1 and p_1 (zone 1) and at T_2 and p_2 (zone 2), respectively. Under the equilibrium conditions, the phase rule is observed in each zone:

☆ Expanded and updated by O.D. Neikov.

$$G = N - \varphi + 2 \quad (5.1)$$

where G , φ , and N are the number of components, phases, and degrees of freedom, respectively.

In the first zone, the vapor of a pure metal consists of a single component and the system possesses two degrees of freedom (the system is bivariant); T and p change independently, and no formation of a new phase takes place. In the second zone, a new liquid or solid phase is precipitated from the vapor. As a result, the number of degrees of freedom decreases to one, the system becomes monovariant, and changing T automatically leads to changes in p , according to the dependence of equilibrium vapor pressure (p_e) on T known for each substance. Changing T to the value at which the liquid phase begins to solidify or the solid phase begins to melt makes the system invariant: it exists only at one temperature and one pressure—the so-called “triple point.” Adding a new component brings in a new independent variable, namely the composition, and the system now acquires one more degree of freedom. The phase rule is observed when there is dissociation of a compound or interaction between several solids during evaporation. In this case, the “reaction pressure” characterizes the system. In the case of the condensed phases, the system is also monovariant.

The formation of a new condensed phase in zone 2 is a consequence of a certain degree of vapor supersaturation (S) occurring either due to decreasing the vapor temperature or to its decomposition:

$$S = \frac{p_1(T_1)}{p_e(T_2)} \quad (5.2)$$

where T and p are bound by the correlation of the Clausius-Clapeyron equation $p_e(T) = p_0 \exp(-E/T)$ with $E = Q/R$; Q is the molar heat of evaporation; and R is the universal gas constant. The value of S depends on the degree of the vapor saturation; $S = 1.0$ for saturated vapors, $S > 1.0$ for super-saturated vapors, and $S < 1.0$ for nonsaturated vapors.

Eq. (5.3) gives the probability of the development of a new phase nucleus in a unit volume of the supersaturated vapor

$$J = K e^{-\Delta G/kT} \quad (5.3)$$

where K is a coefficient; k is the Boltzmann constant; and $\Delta G = 16\pi M^2 \sigma^3 / 3\rho^2 \times k^2 T^2 \ln^2 S$ displays the work of the formation of a condensed phase nucleus (M is the molecular mass of the vapor, σ is the surface tension of the nucleus drop, and ρ is the vapor density).

The physical state of the new phase can be controlled via the degree of vapor supersaturation, $S = p_1(T_1)/p_2(T_2)$, to yield either a monolith or a disperse phase. With $S > 1$, the vapor is supersaturated and is ready for condensation, a quite definite critical value of the degree of supersaturation existing for each substance. The critical supersaturation S_{cr} generated in zone 2 (for metals $S_{cr} > 10^{20}$) is instantaneously realized in building up complexes of vapor molecules – clusters that grow to the critical nuclei of the new condensed phase having its own structure.

In the case of chemical precipitation from the gas phase, an analogous parameter exists, S_{cr} that connects the vapor pressure of the substance in the gas phase with the equilibrium vapor pressure of the metal.

The Hertz-Knudsen equation (Eq. 5.4) describes the rate of substance evaporation and vapor condensation:

$$J = \frac{\alpha(p - p_e)}{(2\pi MRT)^{0.5}} \quad (5.4)$$

where α is either the coefficient of evaporation depending on the nature of the substance and on the properties of the surface or, in the case of precipitation from vapor, the coefficient of condensation (the portion in the total flux of the vapor molecules condensing under given conditions) [3].

The value of the coefficient of evaporation is close to unity in the case of a clean surface of evaporation while the condensation coefficient varies from 10^{-6} to 1.0. In the case when the evaporation-condensation process strongly depends on the total pressure, Eq. (5.5) assumes the form of a simple Langmuir diffusion equation:

$$J = \frac{\alpha D(p - p_e)}{L \Delta T} \quad (5.5)$$

where D is the diffusion coefficient; Δ is the mean free path of the vapor molecules; and L is the thickness of the diffusion layer adjacent to the surface of evaporation or condensation. Eq. (5.5) describes the rate of transfer of vapor molecules when that rate is determined by molecular diffusion [4].

MECHANISMS OF POWDER FORMATION DURING VAPOR CONDENSATION

The following conditions are necessary for the condensation of metallic vapor and the formation of a particulate phase:

1. High supersaturation of the vapor sufficient to trigger simultaneous and multiple formation of nucleation centers.
2. The presence of a neutral gas acting as a medium preventing the coagulation and growth of particles due to diffusion. The presence of foreign inclusions, ions, and chemically active compounds in the neutral gas promotes the nucleation. Forced or natural circulation of the gas ensures the transfer of the condensation heat like the particles and contributes to a variety of forms of their growth.

The formation of powders calls for a molecular-viscous or a viscous flow state of the gas medium (Knudsen number $Kn < 0.01$ and the pressure between 0.01 Pa and 100 kPa). Decreasing the temperature of powder particles, removing the particles from the zone of condensation, passivation of the surface, and dilution of the vapor add to the stability of the disperse system and enable control of particle growth.

The condensation of the metallic vapor and the formation of powder can take place on a cooled surface (a heterophase process), in the bulk gas volume (a homophase process), and as a mixed homoheterophase process as well. The interval of the vapor supersaturation of 10^4 – 10^8 is assumed as a boundary between the first and second processes. At supersaturation values below 10^4 , the formation of the powder occurs on the surface whereas at values above 10^8 , it occurs in the volume; within this interval, a mixed process often takes place.

The formation and properties of the particulate system depend on the heat and mass transfer in the condensation zone. The driving force of the process, that is, the temperature difference between the initial and the final state of the vapor, results from the joint effect of the boundary conditions and the heat evolution during condensation.

Powder formation on a cooled surface (a heterophase process) takes place as a mass growth of crystalline particles, which are nucleated on macroscopic defects of the surface. First, there is formed a multiplicity of little “islands” growing perpendicularly to the substrate along the *C*-axis of the crystals. The islands are shaped as prisms and pyramids of up to 10–20 μm. Then, a second front of crystals is formed with crystals having a *C*-axis parallel to the substrate, and dominant particles are precipitated, intergrowing each other. The new generation of more fine particles replenishes theirs. The density and thickness of the precipitate depend on the growth conditions. The precipitate remains particulate until its relative density does not exceed 30% of the solid. The particle size distribution varies from fractions of microns to several millimeters while the particles are in the shape dendrites or platelets, sometimes with a monocrystalline structure.

The heterophase process is characterized by a high rate of heat and mass transfer interactions, specific thermohydrodynamic phenomena occurring on the boundary of the gaseous and particulate phases, and the formation of intermediate particulate structures.

Different conditions can lead to the formation of powder in the volume (a homophase process), among them adiabatic expansion of the gas medium in the condensation chamber; supercooling of the gas layer adjacent to the intensely cooled surface; feeding cold neutral gas into the vapors; and the presence of chemically active substances able to interact with the gas.

In real processes, almost all the above conditions are present. The particles grow mainly by coagulation, coalescing in the gas flow and precipitating onto cooled surfaces under the influence of thermo- and diffusion-phoretic forces [5,6].

The shape, structure, and size of powder particles depend on the gas dynamics and the heat and mass transfer in the gas phase. At gas temperatures exceeding 0.7 of the metal melting point, particles mostly take a spherical shape, and their formation occurs according to the solid-liquid-solid mode. At temperatures below that value, faceted crystalline shapes appear (prisms, pyramids, and more complex shapes) according to the vapor-crystal mode, with the particles sometimes having a monocrystalline structure. The lower the pressure and vapor concentration, the finer the size of the particles formed. Varying the methods and conditions of condensation allows the production of powders with particle sizes ranging from 0.01 to 50 μm.

Three major types of volume condensation of metallic vapor are of practical interest and can be singled out:

- (1) Diffusion condensation when zones 1 and 2 are combined and evaporation occurs over the open surface of the melted metal.
- (2) Condensation in flame when zone 1 and zone 2 are separated and the vapor enters zone 2 through an orifice, condensing by the action of convection currents.
- (3) Condensation in a vapor jet.

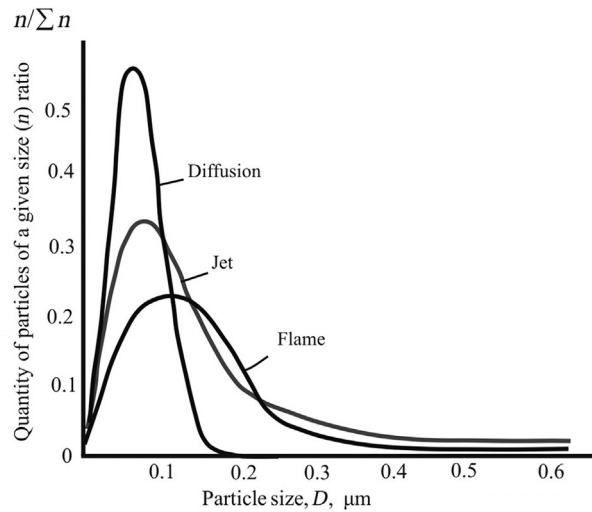
The condensation type is specified by the values of the parameters involved in the equation (Eq. 5.6):

$$J = A/p_{vg}, \quad (5.6)$$

TABLE 5.1 The Condensation Parameters

The Condensation Type	Partial Pressure of the Metal Vapor, p_i	The Degree Indicator, n	The Evaporation/Condensation rate, J , kg/($m^2 \cdot s$)	The Gas Feeding Rate, v , m/s	The Value of the Reynolds number (Re)	The Value of the Richardson Number Criterion for the Stability of the Flow (Ri)
Diffusion	$\ll p_{vg}$	1	≤ 0.001	...	...	...
In-flame	$\leq p_{vg}$	2–4	0.001–0.1	1–5	< 30	> 1
In-jet	$\geq p_{vg}$	4–15	0.1–1.0	5–15	500–1000	< 1

Note: p_{vg} is the total vapor-gas pressure; n is degree of vapor saturation values; $Re = \frac{\rho u L}{\mu}$; $Ri = \frac{g \beta (T_{hot} - T_m) L}{u^2}$ where ρ is the density of the fluid; u is the characteristic velocity of the fluid; L is the characteristic length; μ is the kinematic viscosity; g is the gravitational acceleration; T_{hot} is the wall temperature; and T_m is the average temperature value.

**FIG. 5.1** Particle size number basis distribution of copper powder in diffusion, jet, and flame condensation regimes.

that describes the relation between the rate of evaporation (condensation), J , and the total pressure of the vapor-gas medium, p_{vg} (see Table 5.1). In the last two cases, the values of the current temperature along the jet (flame) axis are inversely proportional to the apparatus length.

The jet (flame) exhibits three zones: the metallic vapor zone (near the jet inlet), the vapor condensation zone, and the aerosol zone. The jet parameters and linear dimensions of the apparatus determine the ratio between zones.

The powder particle size distribution at volume condensation is determined by the rate and mechanism of the growth of condensation nuclei. Particle size distributions in diffusion, flame, and jet modes do not differ substantially, as is shown in Fig. 5.1.

A dominant feature in the diffusion regime is that powder particles increase in size, owing to the vapor condensing on them. Particle size distribution is close to normal while in the flame, the distribution is lognormal. Particle growth is controlled by Brownian coagulation on the free-molecular model. Powder formation in the jet occupies an intermediate position.

The crystal morphology of particles depends on the growth conditions, the properties of the metal, and on the rate of heat and mass transfer processes in the gas medium. Spherical particles form, for the most part, by the vapor-liquid-solid mode and are typical of low-melting metals (tin, zinc, cadmium, and lead). Faceted crystalline forms are produced by the vapor-solid mode (manganese) [5,7].

At certain combinations of the process parameters (surface temperature, volume geometry and dimensions, amount of vapor and gas, etc.), the formation of powder in the bulk may be accompanied by the growth of particles on the cooled

surface. Powders obtained by a mixed process are bimodal and contain particles of mixed shapes and structures, including faceted, spherical, and poly- and monocrystalline. Their dimensions range from tens of nanometers to several millimeters. Ultrafine particles are formed in the bulk while large ones are formed on the surface. Like the heterophase process, the process conditions control the powder granulometric composition and the maximum particle size.

The rate of the metal powder formation obeys the equation (Eq. 5.7):

$$W = KF\rho X^{n+1} \quad (5.7)$$

where F is the geometric factor determining the mass flow to a unit of the volume section; n is the order of the reaction; $X = p_v/p$; p_v is the partial pressure of the vapor in the mixture with a total pressure of p ; and ρ is defined as

$$\rho = \frac{X}{X + (1 - X)M_g/M_v} \quad (5.8)$$

M_g and M_v are molecular weights of the gas and vapor, respectively; K is the process rate constant.

Typical constant values are $K < 0.3$ and $n = 1/2$ for volume condensation; $K > 0.6$ and $n > 1$ for surface condensation; and $0.3 < K < 0.6$ and $n = 1/2$ for the mixed process. The latter value of the parameter n is specific to chemical chain reactions with a quadratic breakage of chains. Parallel to these reactions, the neutral gas initiates the process at $p < 1$ kPa and inhibits it at higher pressures.

During volume condensation, the empirical activation energy values turn out to be lower than the corresponding calculated theoretical values, whereas at surface condensation, on the contrary, they are higher than the theoretical values; in the case of a mixed process, the values almost coincide.

In the process of condensation of metal vapors, the vapor-solid phase transition is accompanied by a rise in temperature due to the release of the internal heat of crystallization ΔH_c , the value of which for metals is significantly high (kJ/kg): Zn – 1884; Mn – 4618; Mg – 5652; and Al – 9211. Empirical equations exist, displaying the interrelations of the basic dimensionless criteria of heat transfer (Nu), mass flow (Re), peculiarities of the gaseous medium (Pr , Sc and Le criteria), and of the vapor-solid phase transition process (Kutateladze criterion) [8].

Intensive heat and mass transfer interactions occurring at the gas-solid boundary, at the front of the growth of the particulate phase, lead to superheating the adjacent gas layer that, in turn, results in a considerable difference between theoretical and empirical values of the coefficients of heat and mass transfer.

The mechanism of heat and mass transfer processes, for example, for the heterophase process of powder formation, can be assumed to be the following. In the gas phase, convective streams and turbulent flows taking place at the growth front of the disperse precipitate enable the heat transfer so that the average temperature of the gas near the boundary becomes higher than the temperature of the metal powder particles. In the boundary region, a temperature gradient arises, growing as the dilution of the medium and the rate of condensation increase (Fig. 5.2).

If T_g is the temperature of the gas of the phase boundary, $p_v(T_g)$ is the equilibrium vapor pressure in the mixture, and T_c , $p_v(T_c)$ are the temperature and the equilibrium vapor pressure at the front of crystal growth, respectively, then the driving force of the mass transfer depends on $\Delta T_1 = (T_g - T_c)$ and $\Delta p = p_v(T_g) - p_v(T_c)$. In this case, the gas temperature in the layer

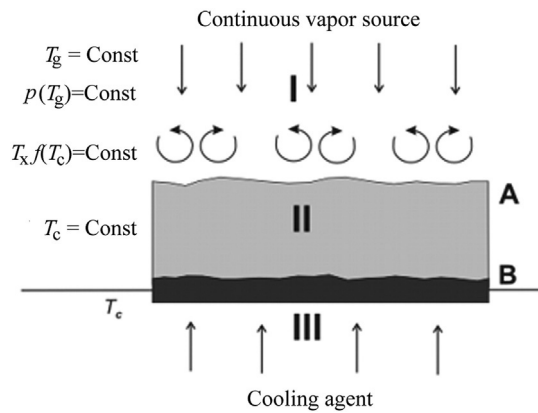
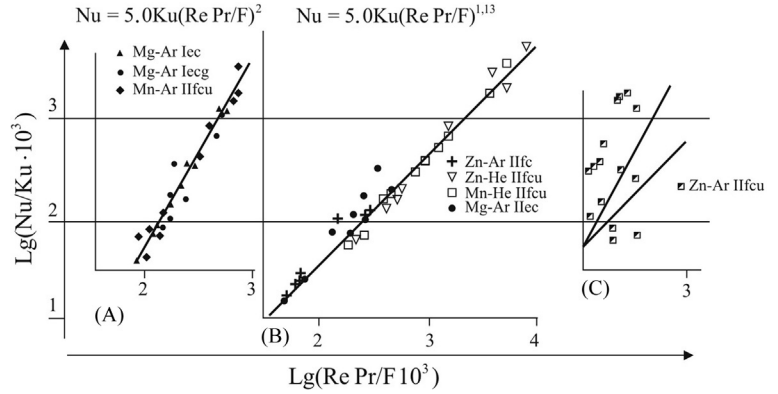


FIG. 5.2 Scheme of heat and mass transfer during crystallization: I is the gas solution region; II is crystal deposit; and III is cooling agent. Designation of the parameters: T_g is the gas temperature; p is the gas pressure. At the boundary A between II and I: T_x is the gas temperature; T_c is the crystal temperature; and the vapor pressure is $p = f(T_c)$. The temperature of the condensation surface B equals T_c .

FIG. 5.3 The general criteria interrelations for (A) heterophase, (B) homophase, and (C) heterohomophase processes: I and II are the first and the second series of experiments; *ec* is eliminated thermoconvection; *ecg* is eliminated thermoconvection and varying geometry of the condensation volume; *fc* is free convection; *fcu* is free convection in an unlimited volume; Mg-Ar is condensation of magnesium vapor in argon; and Zn-He is condensation of zinc vapor in helium.



adjacent to the front of the crystal growth is $T_x > T_c$ and the driving force for the heat transfer is $\Delta T_2 = (T_g - T_x)$. The total heat flow through the phase boundary from the first region into the second is given by

$$\alpha \Delta T_1 = \Delta HW + Wc_p \left[\frac{1}{X\mu p_v} - 1 \right] \Delta T_2 \quad (5.9)$$

The first member of Eq. (5.9) represents the heat of the phase transition; the second one is the physical heat of the gas at the total coefficient of heat exchange α , where the equation for the mixture flow is replaced by the relations:

$$\frac{WM}{XM_v} \quad \text{and} \quad \mu p_v = \frac{M_v}{M_v X + M_g(1-X)};$$

the rest of symbol meanings are given above (Eqs. 5.7 and 5.8).

An analogous equation (Eq. 5.10) for heat and mass transfer at vapor condensation follows from the assumed transfer model:

$$\alpha = \beta K c_p Le^{2/3} \sigma \quad (5.10)$$

or

$$\alpha = Wc_p Ku \quad (5.11)$$

where β is the mass transfer coefficient; K is the Ackerman's process rate correction [9]; c_p is the specific heat capacity of the vapor-gas mixture; Le is the Lewis criterion for the vapor-gas mixture; σ is a dimensionless parameter displaying the effect of the thermohydrodynamic interactions at the phase boundary; $Ku = \Delta H / c_p T_m$, is the Kutateladze phase transition criterion; and T_m is the average temperature value;

$$\sigma = \frac{\Delta p_{\Delta T_1}}{\Delta p_{\Delta T_2}}$$

where $\Delta p_{\Delta T_1}$ is the difference of partial pressures at T_g and T_c temperatures; and $\Delta p_{\Delta T_2}$ is difference of partial pressures at T_g and T_x temperatures.

General criterion equations (Eqs. 5.12 and 5.13) displaying the interrelations of the basic criteria were obtained by processing the empirical data for condensation of metal vapors in various conditions (Fig. 5.3):

$$Nu = AKu(RePr/F)^B \quad (5.12)$$

where F is the geometrical criterion, reflecting the effect of the geometry of the condensation volume; and A and B are empirical constants, calculated as 0.6 and 0.5 for homophase processes, and as 1.1 and 2.0 for heterophase processes, respectively.

In the case of forced convection at a flow rate of 0.001–0.1 m/s, the above equation assumes the form of

$$Nu = A \frac{M_v}{M_g} \left(\frac{RePr}{F} \right)^B \quad (5.13)$$

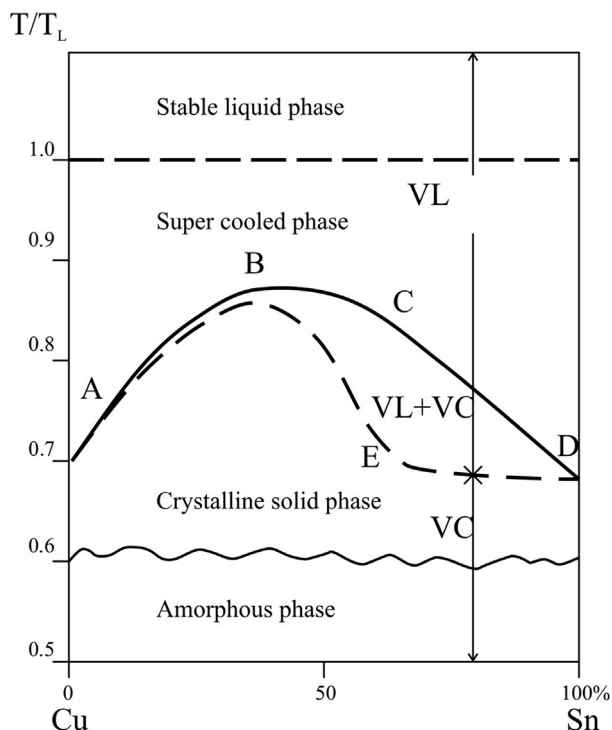


FIG. 5.4 Phase equilibrium diagram of the copper-tin alloy: VL is the vapor-liquid; VC is the crystal-crystal; ABCDE is the transitory (intermediate) region; and T_L is the temperature of the liquidus (the boundary between the vapor and liquid phases).

PECULIARITIES OF THE FORMATION OF ALLOY POWDERS

If the vapor-gas mixture contains several metals at a given stoichiometric ratio, a powder of the corresponding alloy is produced. This can normally be achieved either by evaporating an alloy of required composition, by evaporation of the components of the alloy separately from independent sources, or by coprecipitation of powders produced by decomposition of a mixture of metal-containing gaseous compounds.

The composition and properties of the final powder depend mostly on the type of the corresponding phase diagram. The formation of a mixture of particles of individual components is a feature of eutectic alloys with the mixing energy $U > 0$ (Pb – Sb). For the alloys with the cigar-shaped phase diagram with $U = 0$ (Bi – Sb, Mg – Cd), a joint condensation of metals takes place on common nuclei, and the resulting powder consists of particles of the alloy. In the case of metals forming congruent and incongruent melting of the compounds with $U < 0$ (Cd – Sb, Pb – Bi), the powder often comprises particles of intermetallic compounds.

The formation of alloy powders displays peculiarities associated with their phase diagram. The latter determines the composition and structure of the surface layer of the evaporating melt and, therefore, the properties of powders. In cases where the alloy constituents form no chemical compounds, the knowledge of the phase diagram suffices to predict those properties. For alloys displaying strong particle interaction whose components form a number of intermetallic compounds, vapor and melt composition dependent on the temperature and pressure values obey the Vrevsky laws. It has been found for such alloys that S_{cr} approaches unity due to the presence of ready cluster groups (condensation nuclei) in the vapor.

The rate J of evaporation (condensation) of an n -component alloy obeys the equation:

$$J = \sum_1^n J_i(a_i b_i)$$

where a is the activity of the alloy component and b is the mole fraction of clusters in the vapor. The latter can be evaluated correctly enough from the model of ideal associated solutions.

Fig. 5.4 shows the phase equilibrium diagram in the condensation process of a copper-tin alloy.

Using the temperature as a control parameter enables the production of either spherical (“super cooled phase,” Fig. 5.5A), faceted crystalline particles (region “crystalline solid phase,” Fig. 5.5B), or amorphous powder (Fig. 5.5C) [7].

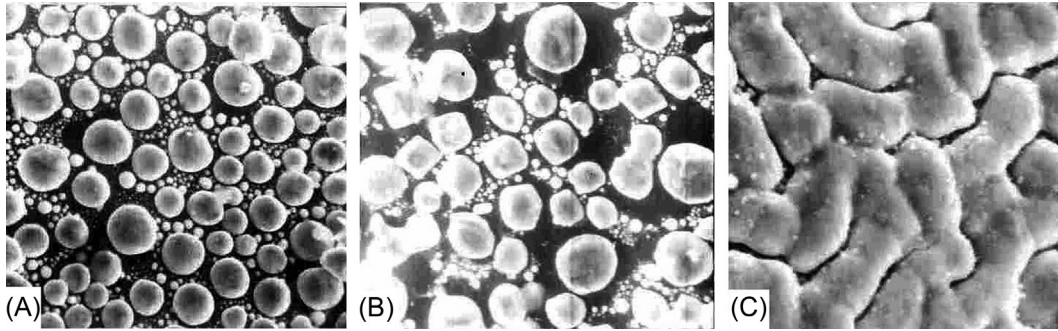


FIG. 5.5 Micrographs of the copper-tin alloy powders obtained in the condensation process in several regions of the phase equilibrium diagram (see Fig. 5.4): (A), VL, vapor-liquid $\times 1200$; (B), VC, crystal-crystal $\times 2500$; (C), ABCDE $\times 500$.

CONTROLLING PRINCIPLES OF THE CONDENSATION PROCESS

The theoretical concepts given above form a foundation for the development of the basic IGC technology for production of metal powders. The condensation process is carried out by one of three modes, depending of the targeted powder properties:

- In the bulk of the condensation zone at the starting supersaturation degree of the vapor $S_{cr} > 10^6$.
- On a cooled surface placed in the condensation zone at $S_{cr} < 10^6$.
- In a mixed process, simultaneously in the volume and on the cooled surface. In the absence of thermoconvection, the temperature gradients can be evaluated using the equation

$$T = T_2(T_2 - T_1) \frac{e^B - 1}{e^b - 1}$$

$$\text{with } b = \frac{Q_v C_v (r_2 - r_1)}{\lambda 2\pi r_1 r_2 (1 - \cos \alpha)} \text{ and } B = \frac{b r_1}{r}.$$

where b and B are distances between the vapor source and the powder receiver with crossing angle α ; Q_v and C_v are the flow and the specific heat capacity of the vapor; r_1 , r_2 are the radii of a cell at the source and the receiver levels; r is the radius of the cell when the latter has the shape of a sphere; and λ is the heat conductivity coefficient.

The concentration and vapor supersaturation gradients are determined by differential equations representing the interrelations of the parameters of the vapor-gas medium (density, viscosity, partial pressure, etc.) and the geometrical characteristics of the condensation volume (shape, linear dimensions, distance from the vapor source to the crystallization surface, etc.).

The control of the powder dispersivity is realized through the residence time of the powder in the condensation zone and by selecting appropriate values of the condensation parameters. Quick cooling of the vapor and a short residence time in the condensation zone result in the formation of fine and ultrafine powders while full or partial condensation on the cooled surface gives more coarse powders with the particle size in the range from several microns to several millimeters.

In the case of bulk condensation, the control of the shape and structure of particles is realized through selecting such boundary conditions and the rate of heat transfer, which can provide the condensation process going in either vapor-solid or vapor-liquid-solid mode. In the first case, faceted monocrystalline particles are formed while, in the other case, particles of a spherical shape and semicrystalline structure are formed. In the case of condensation on the surface, particles of faceted shapes and of mono-or polycrystalline structure prevail.

Steady boundary conditions and permanent geometrical parameters of the condensation zone provide stable quality characteristics of the powder. The criterion equations (Eqs. 5.12 and 5.13) are helpful for designing the equipment and selecting the process parameters.

POWDER PRODUCTION AND APPLICATION OF GAS-PHASE PROCESSES

The gas-phase methods find wide application for the manufacture of semiproducts directly from metal vapor, by omittance the operation of nanopowder obtaining. The simultaneous evaporation of several materials, mixing their vapor flows, followed by condensation on a substrate and subsequent thermomechanical treatment, make it possible to design materials with unique properties.

Physical Vapor Deposition (PVD) Process

The PVD process includes melting the metal; producing a vapor; feeding the vapor into the zone of condensation; condensing the vapor into powder either in the free space or on a cooled surface (in a static or moving gas medium); and accumulating the powder in the cold part of the volume. The process is carried out in the presence of a neutral gas, sometimes with the addition of an oxidant or, on the contrary, of a passivating agent.

In the PVD process installations, a source material is transformed from a solid to a gaseous state by the technique, which may be divided into two physical process groups: evaporation and physical sputtering [10]. The evaporation is realized by means of resistance or induction furnaces, electron beams, low-voltage arcs, hollow cathodes, or laser beam heating. The processes can be carried out with additional ionization or without, in a reactant gas medium or without, and with bias voltage or without [11,12]. Diode or magnetron sputtering is used by direct current or rf current in the reactant gas medium or without, with bias voltage or without, or with additional modification of the magnetic field or without [13–15].

A zinc powder of high purity is manufactured in plate-type rectifying columns. The zinc vapor is fed via vapor pipes into one or several large tanks with water-cooled walls that act as condensers. A neutral gas is fed into the column. The lower part of the column is equipped with a bunker for accumulation of the powder and with an appliance for its discharge. The discharged powder is subjected to classification. Typically, the output of columns and condensers is within the range of 200–400 kg/h; the rejected fractions are normally recycled. The temperature of the column controls the powder quality (see also Chapter 20).

There are different types of equipment for the laboratory and pilot plant scale intended for the production of powders by the evaporation-condensation method. As a rule, an apparatus is fitted with an induction or resistance heated evaporator and moving cooled surfaces (drum, belt, or disc) for vapor condensation. Sometimes a common volume is used both for evaporation and condensation. The nature of the metal governs the process. The manufacturing of volatile metal powders (zinc, magnesium, cadmium, lead) calls for atmospheric pressure while aluminum and copper powders are produced at a considerably reduced pressure of a neutral gas. The lower the pressure, the lower the evaporator temperature.

One of the installations for zinc powder manufacturing is shown in Fig. 5.6. A vertical apparatus of $>10\text{ m}^3$ in volume equipped with a sealed water-cooled evaporated chamber consists of several sections, each up to 1 m high, made so as to permit the regulation of the volume of the condensation zone.

Each of the sections includes a 100 kW inductor and a graphite crucible with a capacity of up to 300 kg of molten zinc. The design of the crucible facilitates feeding nitrogen into the condensation zone. The zinc vapors are directed by the nitrogen flow toward a rotating water-cooled drum condenser. The powder deposited on the drum surface is cut away with a special blade and dropped into a bunker. The unit operates at the evaporator temperature of 923–1073 K in rarefied atmosphere at 3–20 kPa. The unit yield is 10–15 kg/h of ultrafine zinc.

A modular type apparatus named “Tuman” (trade name “Fog”) was developed at the Institute of Metallurgy, Ural Branch of the Russian Academy of Science. It can feature either combined or separate zones of evaporation and condensation and is less sophisticated in design and easier in operation in comparison with the above installation.

The Tuman unit (Fig. 5.7) has an annual output of 120 tons of zinc powder. Electric power consumption is up to 40 kW; the operational cycle of 25 days is followed by a one-day stop for maintenance. The machine is easy to operate and to

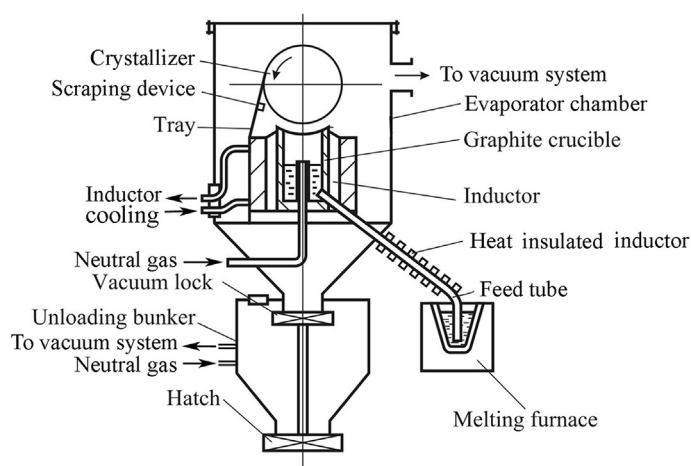


FIG. 5.6 An apparatus for the production of ultrafine powders.

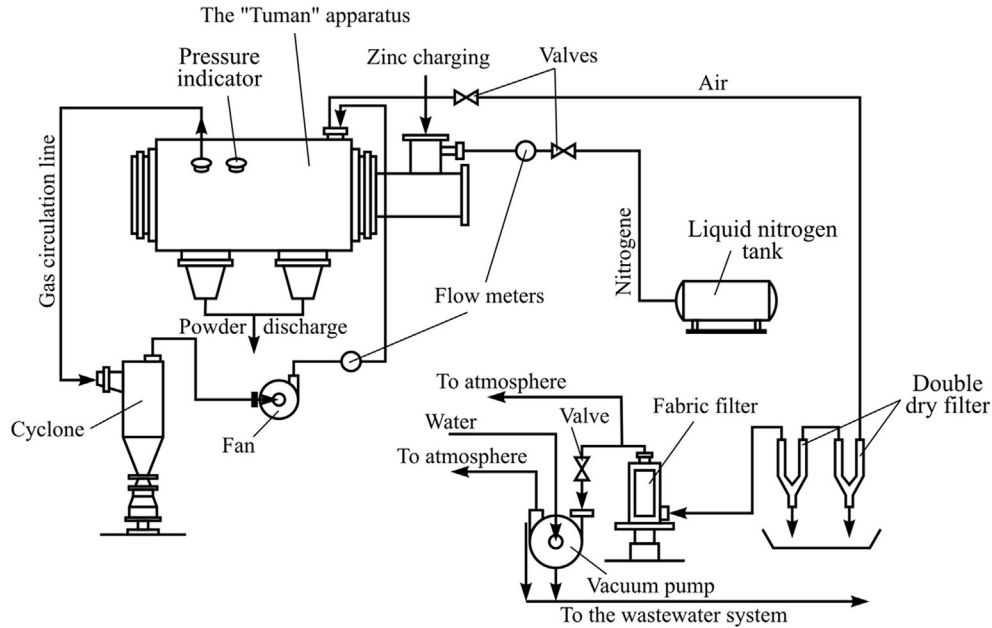


FIG. 5.7 The schematic drawing of the “Tuman” (“Fog”) unit.

TABLE 5.2 Specification of Metal Powders Produced by the PVD Technique

The Powder Mark (as Given by the Manufacturer)	Active Metal Content, % (min)	Bulk Density (g/cm ³)	Specific Surface (m ² /g)	Average Particle Size (μm)
Zinc powder PZHD	96.0	1.4–2.0	0.1–0.6	4–12
Copper powder PCND	99.98	0.4–1.3	8.0–12.0	0.1–0.3
Bronze powder PBND	Cu 89.0–98.0; Sn 1.0–10.0 (typical)	0.3–1.2	3.0–8.0	0.1–0.2
Lead powder PLHD	94.0–95.0	1.8–3.6	...	0.5–3.0

convert from the production of one type of powder to another. The controllable process ensures the production of metal powders with a targeted particle size in the micron, submicron, and nanometer ranges with high product yields.

Commercial scale Tuman units are in use at the VMP R&D Company’s plant in Yekaterinburg (Russia) for manufacturing fine and ultrafine powders of nonferrous metals and alloys (Table 5.2) (see also Chapter 20).

Properties of Powders and Areas of their Usage

Powders produced by the evaporation-condensation process may be divided into two categories according to their particle size: ultrafine (micron and submicron) and nanosized powders, with the categories differing radically in their chemical and physical properties. The nanopowders display higher activity due to their high specific surface area and a peculiar “nonequilibrium” structure of particles (see Fig. 5.8 and Table 5.1). This makes them fit for such specific fields as the production of rocket fuel, catalysts, heat absorbents, and microelectronics. The properties of ultrafine powders are more ordinary, and the range of their application is much wider, from pigments for paints and varnishes to items of powder metallurgy. Nanopowders of copper and its alloys have an average particle size of 100–200 nm. Areas of applications of these powders are considered in more detail in Chapter 19.

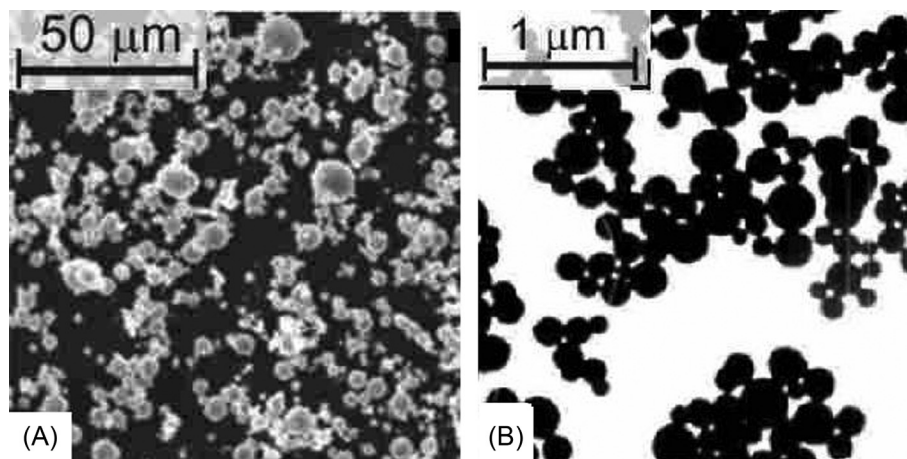


FIG. 5.8 Micrographs of fine (A) and ultrafine (B) powders produced by the PVD technique.

Chemical Vapor Deposition (CVD) Process

CVD is practiced in a variety of formats. These processes generally differ in the means by which chemical reactions are initiated. Those classified by operating pressure:

- Atmospheric pressure CVD (APCVD)-CVD at atmospheric pressure.
- Low-pressure CVD (LPCVD)-CVD at subatmospheric pressures. Reduced pressures tend to reduce unwanted gas-phase reactions and improve film uniformity across the wafer.
- Ultrahigh vacuum CVD (UHVCVD)-CVD at very low pressure, typically below 10^{-6} Pa ($\sim 10^{-8}$ torr). Note that in other fields, a lower division between high and **ultrahigh vacuum** is common, often 10^{-7} Pa.

Most modern CVD processes are either LPCVD or UHVCVD.

Those classified by the physical characteristics of vapor:

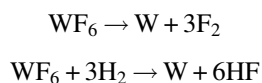
- Aerosol-assisted CVD (AACVD)-CVD in which the precursors are transported to the substrate by means of a liquid/gas aerosol, which can be generated ultrasonically. This technique is suitable for use with nonvolatile precursors.
- Direct liquid injection CVD (DLICVD)-CVD in which the precursors are in liquid form (liquid or solid dissolved in a convenient solvent). Liquid solutions are injected in a vaporization chamber toward injectors. The precursor vapors are then transported to the substrate, as in classical CVD. This technique is suitable for use on liquid or solid precursors. High growth rates can be reached using this technique.

Plasma methods (also cooled **plasma processing**):

- Microwave plasma-assisted CVD (MPCVD).
- **Plasma-Enhanced CVD** (PECVD)-CVD that utilizes **plasma** to enhance the chemical reaction rates of the precursors. PECVD processing allows deposition at lower temperatures, which is often critical in the manufacture of semiconductors. The lower temperatures also allow for the deposition of organic coatings, such as plasma polymers, that have been used for nanoparticle surface functionalization.
- Remote plasma-enhanced CVD (RPECVD)—similar to PECVD except that the wafer substrate is not directly in the plasma discharge region. Removing the wafer from the plasma region allows processing temperatures down to room temperature.

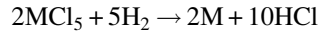
In typical CVD, the substrate is exposed to one or more **volatile precursors** in possession of high vapor pressure by low temperatures (373–673 K), such as chlorides and organometallic compounds, which **react** and/or **decompose** on the substrate surface to produce the desired deposit. Frequently, volatile **by-products** are also produced, which are removed by gas flow through the reaction chamber.

CVD for **tungsten** is achieved from **tungsten hexafluoride** (WF_6), which may be deposited in two ways:

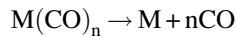


Other metals, especially aluminum and copper, can be deposited by CVD. $\text{Cu}(\text{hfac})_2$ is acceptable as a precursor for copper CVD [14]. Copper is typically deposited by electroplating. The precursors for copper chemical vapor deposition are discussed in the work [16]. Aluminum can be deposited from triisobutylaluminum (TIBAL) and related organoaluminum compounds.

CVD for titanium, nickel, and refractory metals, including molybdenum, tantalum, niobium, and rhenium, is widely used [17–19]. These metals can form useful silicides when deposited onto silicon. Mo, Ta, and Ti are deposited by LPCVD from their pentachlorides. Nickel, molybdenum, and tungsten can be deposited at low temperatures from their carbonyl precursors. In general, for an arbitrary metal M, the chloride deposition reaction is as follows:

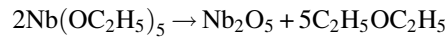


whereas the carbonyl decomposition reaction can happen spontaneously under thermal treatment or acoustic cavitation and is as follows:



The decomposition of metal carbonyls is often violently precipitated by moisture or air, where oxygen reacts with the metal precursor to form a metal or metal oxide along with carbon dioxide.

Niobium(V) oxide layers can be produced by the thermal decomposition of niobium(V) ethoxide with the loss of diethyl ether [20,21], according to the equation:



Chemical reactions of the CVD processes occur in the immediate vicinity or on the surface of an under-treatment article at temperatures up to 1373 K, what bounds CVD application for materials, such as high-speed steel, which properties are lost consequently of the tempering (Fig. 5.9).

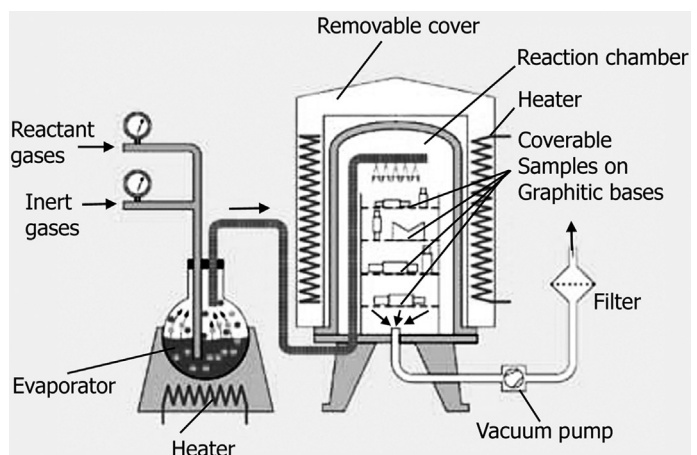
Recently, PECVD modification was developed permitting the decrease of the coating temperature up to temperatures lower than 773 K [22]. In fact, this method corresponds to a combination of two basal methods of gas-phase precipitation, so far as in this CVD modification the coating occurs in the plasma medium, such as by PVD. A layout view of the PECVD parallel plate reactor is represented in Fig. 5.10.

Both the gas-phase methods CVD and PVD are used for wear-resistant and thermostable covering. Microfabrication processes widely use CVD to deposit materials in various forms, including monocrystalline, polycrystalline, amorphous, and epitaxial. These materials include silicon (SiO_2 , germanium, carbide, nitride, oxynitride), carbon (fiber, nanofibers, nanotubes, diamond and graphene), fluorocarbons, filaments, tungsten, titanium nitride, and various high-*k* dielectrics.

Health and Environment Protection

The safety system for the plants operating the evaporation-condensation-based processes must include, apart from the conventional topics relating to the high-temperature and vacuum processes, the issues meeting safety requirements for the flammable and explosive materials. In terms of environmental protection and health, special measures are required such

FIG. 5.9 The schematic drawing of the installation for deposition by the CVD technique.



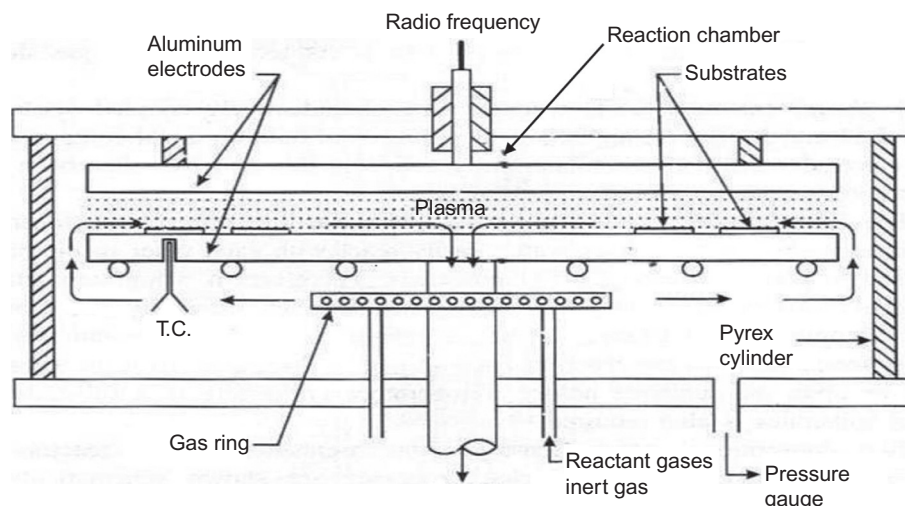


FIG. 5.10 Parallel-plate reactor of the plasma-enhanced CVD process.

as the neutralization of gas and water effluents, cleaning the working areas, providing the workers with the necessary individual protection gear, etc. More specific requirements are described in the chapters dealing with the production of powders of individual metals.

REFERENCES

- [1] Frishberg IV, Kvater LI, Kuz'min BP, Gribovskii SV. The gas-phase method of preparing powders. Moscow: Nauka Publishers; 1978 [In Russian].
- [2] Naboychenko SS, editor. Handbook of NonFerrous Metal Powders. Moscow: Metallurgia Publishers; 1997 [in Russian].
- [3] Brock JR. The dynamics of ultrafine particles: a survey. *J Aerosol Sci* 1979;10(2):192–5.
- [4] Hirsh D, Pound G. Evaporation and condensation. Moscow: Metallurgiya Publishers; 1966 [in Russian].
- [5] Subbotina OY, Kishkoparov NV, Frishberg IV. Genesis of ultrafine metal powders built on their particle size distribution. In: Proceedings of the 5th All-Russian Conf. on Physical Chemistry of Ultra Disperse Systems, Part. Vol. 1. Yekaterinburg (Russia): Publ. the Ural Branch of the Russian Academy of Sciences; 2001. p. 53–7 [in Russian].
- [6] Subbotina OY, Kishkoparov NV, Frishberg IV. Mechanism and kinetics of coagulation of ultrafine metal particles formed at an IGC process. In: Proceedings of the 5th All-Russian Conference on Physical Chemistry of Ultra Disperse Systems, Part 1. Yekaterinburg (Russia): Publ. the Ural Branch of the Russian Academy of Sciences; 2001. p. 58–63 [in Russian].
- [7] Frishberg IV. Gas-phase metallurgy: from an idea to the commercial manufacturing of fine metal powders and products based thereon. In: Proceedings of the EuroPM 1997 International Conference. Munich: Publ. EPMA; 1998. p. 252–4.
- [8] Kutateladze SS. The heat transfer by condensation and boiling. Moscow–Leningrad: Mashgiz Publ; 1952 [in Russian].
- [9] Ackerman G. Wärmeübergang und molekuläre stoffübertragung im gleichen feld bei grossen temperatur- und partialdruck-differenzen. *VDJ-Forschungsh* 1937;382:1–16.
- [10] Fuller L. Physical vapor deposition—evaporation and sputtering. Webpage (26 Nov 2013): <http://people.rit.edu/lffeee/PVD_Recipes.pdf> [Accessed 31 October 2017].
- [11] Harsha KSS. Principles of physical vapor deposition of thin films. Great Britain: Elsevier; 2006.
- [12] Movchan BA. Inorganic materials and coatings produced by EBPED. *Surf Eng* 2006;22(1):35–46.
- [13] Sigmund P. Mechanisms and theory of physical sputtering by particle impact. *Nucl Instrum Meth Phys Res* 1987;27:1–20.
- [14] Kuzmichev AI. Magnetron sputtering systems. Kiev (Ukraine): Avers; 2008 [in Russian].
- [15] Behrisch R. Sputtering by particle bombardment: experiments and computer calculations from threshold to mev energies. Eckstein W., editor. Berlin: Springer; 2007.
- [16] BuFaroosha M. Precursors for copper chemical vapor deposition. Dr dissertation, B.S. United Arab Emirates University, 2002.
- [17] Fortini AJ. Low-cost, high-performance refractory metal alloys for rocket engines. In: Proceedings of the 9th Tungsten, refractory & hardmaterials conference. Orlando (FL, U.S.A.): Metal Powder Industries Federation; 2014. p. 537–43.
- [18] ATI Wah Chang. Technical data sheet C-103 niobium properties and products. Overview the requirement for high-performance. 2017 <<http://www.industrycortex.com/datasheets/profile/2482808.pdf>>.
- [19] Tuffias R, Brockmeyer J, Fortini A, Williams B, Duffy A, Kaplan R. Engineering issues of iridium/rhenium rocket engines revisited. In: Proceedings of the 35th AIAA/ASME/SAE/ASEE joint propulsion conference. Los Angeles (U.S.A.): American Institute of Aeronautics and Astronautics, Reston, VA; 1999. p. 1–7.

- [20] Maruyama T, Konodawa T. Electrochromic properties of niobium oxide thin films prepared by chemical vapor deposition. *J Electrochem Soc* 1994;141(10):2868–71. <https://doi.org/10.1149/1.2059247>.
- [21] Rahtu A. *Atomic layer deposition of high permittivity oxides: film growth and in situ studies*. Academic dissertation. University of Helsinki 2002. ISBN 952-10-0646-3.
- [22] Cardenas P. and Tung D. Plasma enhanced chemical vapor deposition (PECVD). <<http://www.ece.umd.edu/class/enee416.F2007/GroupActivities/Presentation5.pdf>>; 2007 [Accessed 31 January 2018].